Differences among total and in vitro digestible phosphorus content of meat and milk products.

OBJECTIVE: Meat and milk products are important sources of dietary phosphorus (P) and protein. The use of P additives is common both in processed cheese and meat products. Measurement of in vitro digestible phosphorus (DP) content of foods may reflect absorbability of P. The objective of this study was to measure both total phosphorus (TP) and DP contents of selected meat and milk products and to compare amounts of TP and DP and the proportion of DP to TP among different foods. METHODS: TP and DP contents of 21 meat and milk products were measured by inductively coupled plasma optical emission spectrometry (ICP-OES). In DP analysis, samples were digested enzymatically, in principle, in the same way as in the alimentary canal before the analyses. The most popular national brands of meat and milk products were chosen for analysis. RESULTS: The highest TP and DP contents were found in processed and hard cheeses; the lowest, in milk and cottage cheese. TP and DP contents in sausages and cold cuts were lower than those in cheeses. Chicken, pork, beef, and rainbow trout contained similar amounts of TP, but slightly more variation was found in their DP contents. CONCLUSIONS: Foods containing P additives have a high content of DP. Our study confirms that cottage cheese and unenhanced meats are better choices than processed or hard cheeses, sausages, and cold cuts for chronic kidney disease patients, based on their lower P-to-protein ratios and sodium contents. The results support previous findings of better P absorbability in foods of animal origin than in, for example, legumes.